Homelab architecture

Physical hosts, service placement and the shared platform.

People & devices
Browser · phone · workstation
Home LAN or remote Tailscale

Two access paths
Public: Internet → router → Caddy → selected services
Private: LAN / Tailscale → Caddy or native LAN protocols
Tailscale subnet routing runs on all three Proxmox hosts.

Cloudflare
DNS-only records · DNS-01 TLS
Private defaults; public exceptions
DNS is not a traffic tunnel.

slate · external GCP VM
Caddy · Uptime Kuma · ntfy
Independent monitoring & alerts
Checks home over Tailscale

Home router / LAN fabric — Ethernet: 1 Gb/s full duplex verified at px10, px20 and px30 — Wi-Fi → bulb & AC

px10 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · ZFS OS + data pools

dev · VM 100
Codex · Claude Code · Hermes · OpenCode
Herdr · shared tools / MCP · development workloads

observability · LXC 104
Beszel · Grafana · VictoriaMetrics
Uptime Kuma · PVE exporter

store · LXC 109
Samba: AV · BACKUP · ISO
Host USB SSDs → bind mounts → SMB shares

k8s-201 · VM 201
Talos · control plane + workloads
4 vCPU · up to 16 GiB RAM · dedicated 500 GiB data disk

px20 — Dell OptiPlex 5060

i5-8400T / 6 cores · 32 GB RAM · ZFS OS + data pools

ctr · VM 204 / Docker + Portainer
Media Jellyfin · qBittorrent · Motrix
Personal Immich · Karakeep · OpenCloud · Paperless
Notes Nivra · separate friend instance
Build Forgejo · Docker Registry · mirror worker
Automate n8n · SearXNG · isolated sandbox
Agents OpenViking · Ollama · Omni
Helpers PostgreSQL · Valkey · Meilisearch · updater

orva · VM 106
Serverless functions · Quick Share

k8s-202 · VM 202
Talos · control plane + workloads
4 vCPU · up to 16 GiB RAM · dedicated 500 GiB data disk

px30 — Dell OptiPlex 7040

i5-6500TE / 4 cores · 32 GB RAM · ZFS OS + data pools

proxy · LXC 101
Caddy · HTTPS routing / TLS

dns · LXC 102
AdGuard Home → Unbound recursion

auth · LXC 103
Pocket ID / OIDC · Tinyauth / forward-auth

s3 · LXC 105
Cairn · RustFS / S3-compatible storage

haos · VM 107
Home Assistant · Mosquitto · local device control

k8s-203 · VM 203
Talos · control plane + workloads
4 vCPU · up to 16 GiB RAM · dedicated 500 GiB data disk

Placement is a snapshot; managed guests can relocate.

One Kubernetes cluster across all three physical hosts

Talos management: Omni on ctr. Kubernetes desired state: GitHub → Argo CD.

Networking & operations

Cilium CNI · L2 LoadBalancer services · CoreDNS
Argo CD · Headlamp · Homepage · metrics-server
vmagent · kube-state-metrics · node metrics

Persistent storage

Longhorn block volumes on the 500 GiB disks
Current volumes: 2 storage replicas each
Data crosses hosts; replication is not backup.

Supabase

CloudNativePG: 1 primary + 2 hot standbys
Studio · Auth · REST · Realtime · Edge Functions
Storage / imgproxy · Kong · pooler · logs / analytics

Storage boundaries

ZFS → VM/LXC disks and selected 5-minute replicas
USB on px10: 4 TB AV / BACKUP + 256 GB ISO
ctr local data → app state; S3 → Cairn / RustFS objects

Availability boundaries

Proxmox HA manages the listed non-Kubernetes guests.
USB shares need physical-disk recovery after host loss.
DNS, Caddy and several apps remain single instances.

Smart home

Home Assistant → Wipro bulb / Tuya LAN
Home Assistant → Samsung AC / local DTLS
Frigate → camera feeds; HA → viewing / automation

Logical LAN fabric shown; router, switch/AP models and exact cabling have not been audited.

How traffic and data move

The important connections, without drawing every packet or replica.

Web access & identity

Client
LAN / Tailscale for private apps
Internet for selected public apps

→

Caddy · proxy LXC
HTTPS termination → backend routes
Source restrictions per service

→

Application
Native login / API credentials
No blanket public or SSO bypass

Pocket ID + Tinyauth · auth LXC
Native OIDC where supported
Caddy forward-auth for selected browser routes

↑ Identity is a policy dependency; API/MCP routes retain their own authentication.

Name lookup: AdGuard → Unbound

GitOps & cluster operations

GitHub · Homelab repository
Declarative non-secret configuration
Reviewed changes → desired state

⇢

Argo CD · Kubernetes
Reconciles tracked applications
Signed public webhook; private UI

⇢

Talos / Kubernetes
Cilium network · Longhorn volumes
Services scheduled across three VMs

⇠

Omni · ctr
Manages the three Talos machines
Runs on ctr, not inside Kubernetes

Longhorn: storage-level replicas. CloudNativePG: separate PostgreSQL streaming replicas and failover.

Agents & shared memory

dev · four agents
Codex · Claude Code
Hermes · OpenCode

→

OpenViking · ctr
Shared account-scoped memory
Native plugins / provider and MCP

→

Ollama · ctr
Local embedding service
Not the agents' primary chat model

Tools & integrations
Orva · Nivra · Supabase · browser / developer MCPs
Hermes → Karakeep CLI / Home Assistant API

↑ External model APIs supply inference. Telegram / Photon iMessage reach the Hermes gateway on dev.

Media & object storage

Media path
qBittorrent / Motrix → Store AV share → Jellyfin
Encrypted SMB3 directly over the LAN; not through Caddy.
Media stays on Store's USB SSD, not ctr's OS disk.

Temporary sharing
Quick Share CLI → Orva function → Cairn S3
Short Orva share URL → function proxy → object download.
Cairn is a separate native service on the s3 LXC.

Container images
Docker client → Registry → RustFS S3
Registry runs on ctr; image layers live in RustFS.
Cairn and RustFS share one LXC / host failure domain.

Proxmox nodes use Store BACKUP / ISO over SMB directly. App databases remain on their own local or block volumes.

Home automation & visibility

Hermes / user
Home Assistant API / app
Existing authenticated access

→

Home Assistant · haos
Wipro bulb → Tuya LAN
Samsung AC → local DTLS / CoAP

←

Frigate · ctr / Intel iGPU
Camera ingest / recording
RTSP restream + MQTT → HA

Metrics · observability LXC
Exporters + Kubernetes vmagent → VictoriaMetrics
Grafana / HA node views; Beszel / Kuma monitoring

Local device protocols avoid cloud commands; a full WAN-outage acceptance test remains outstanding.

Solid arrows: application / data path → Dashed arrows: configuration / dependency ⇢ Not a live health map; see the inventory notes for verification limits.

Schemas are simplified; replicated storage and HA are not independent backups.